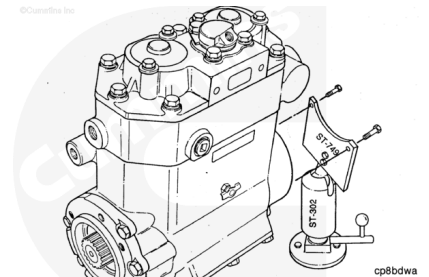


Trainings ▾

Disassemble

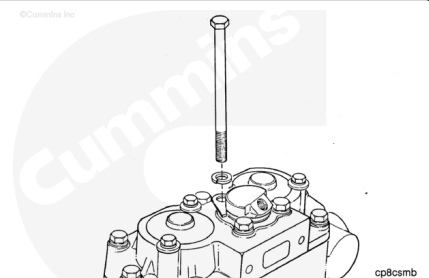
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift component or assembly.

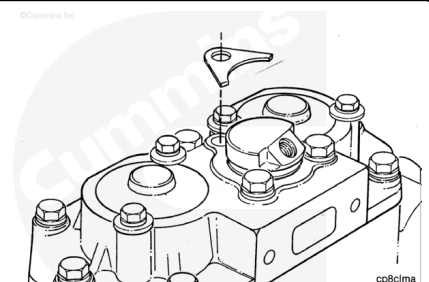


Use two 5/16 - 18 x 1 1/4 inch capscrews to mount the air compressor on the mounting plate, Part Number ST-749, which is used with the ball joint vise, Part Number ST-302.

Remove the center unloader body hexagon head capscrew and lock washer.

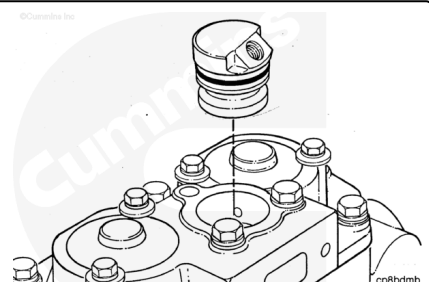


Remove the retaining clamp.



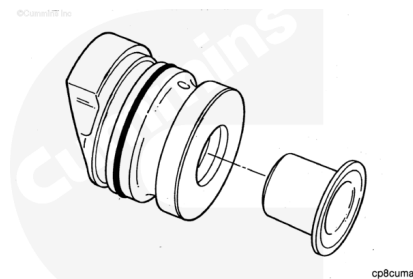
⚠ CAUTION ⚠

The unloader valve body is installed with spring tension. Carefully remove to avoid personal injury. Wear face and eye protection.

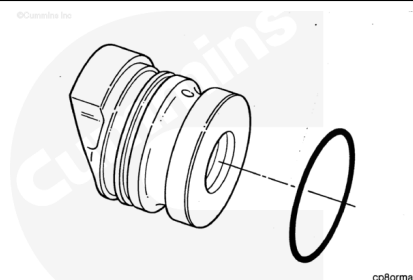


Remove the unloading valve body.

Remove the unloader valve cap, if installed.

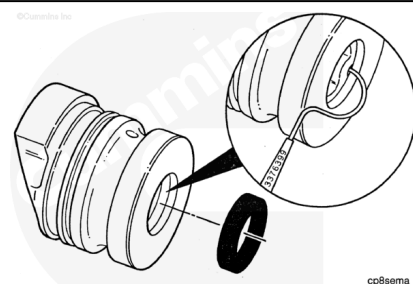


Remove and discard the o-ring seal.

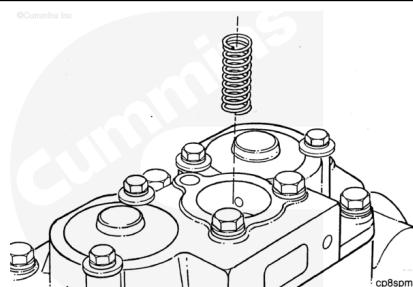


Use an o-ring pick to remove the rectangular ring seal, if installed.

Discard the seal.

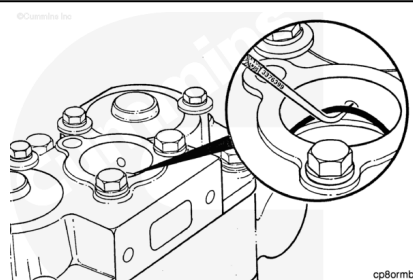


Remove the unloader valve cap spring.

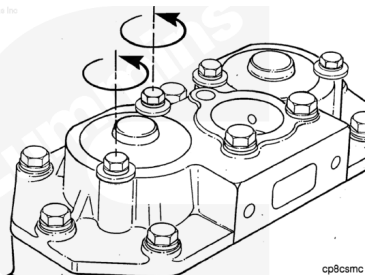


Use an o-ring pick to remove the o-ring seal.

Discard the seal.



Loosen the two unloader body hexagon head capscrews.

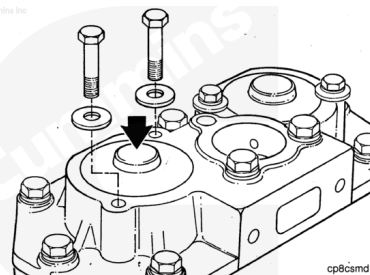


cp8csmc

⚠ WARNING ⚠

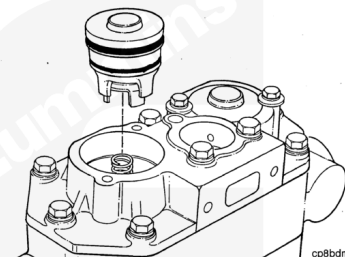
The unloader body is installed with spring tension, it must be removed carefully to avoid personal injury. Always wear protective eye wear.

Hold down on the unloader body and slowly remove the two unloader body capscrews and the two plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



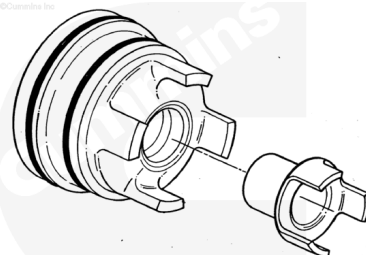
cp8csmc

Remove the unloader valve body.



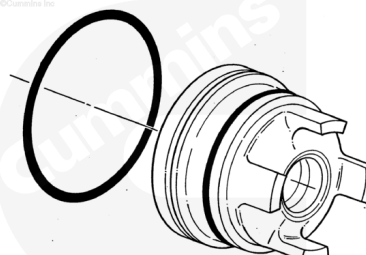
cp8bdme

Remove the unloader cap.



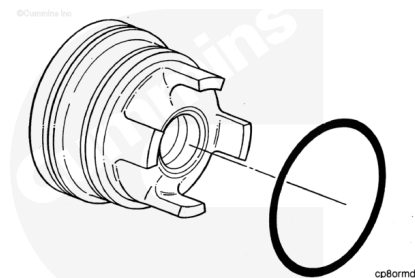
cp8cumb

Remove and discard the top o-ring seal.



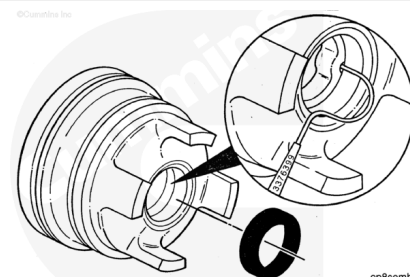
cp8ormc

Remove and discard the bottom o-ring seal.



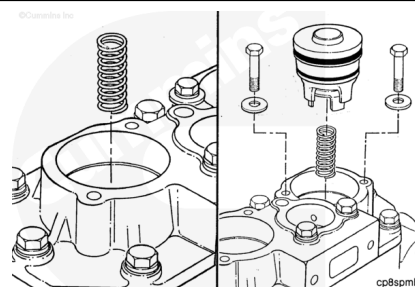
Use an o-ring pick to remove the rectangular ring seal.

Discard the seal.



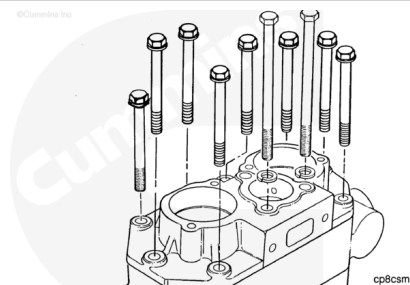
Remove the unloader cap spring.

Repeat the last eight steps to remove the other unloader body assembly.

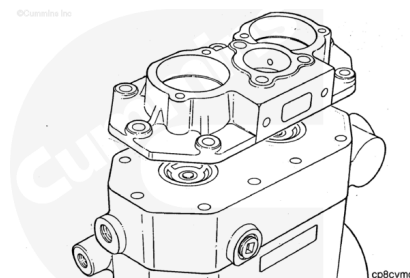


For the ST676, remove the ten remaining hexagon head capscrews and two lock washers.

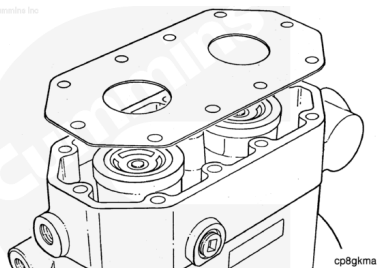
For the ST773, remove the eight captive washer capscrews and the two hexagon head capscrews and lock washers.



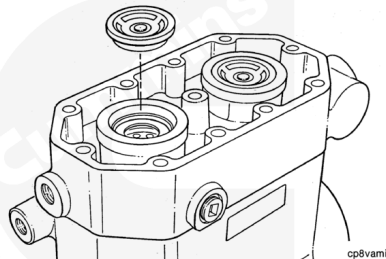
Remove the cover.



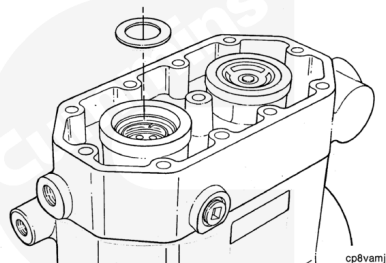
Remove and discard the cover gasket.



Remove the intake valve seat.

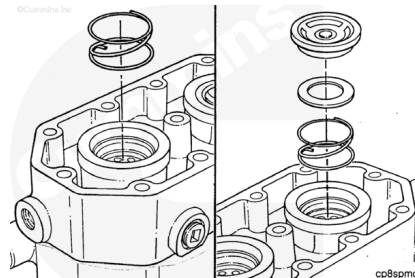


Remove the intake valve.

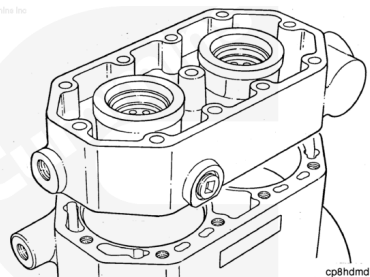


Remove the intake valve spring.

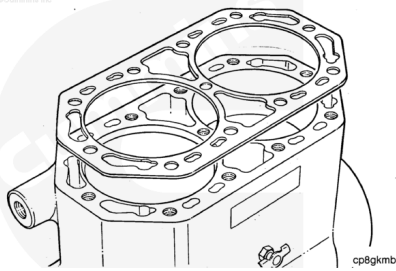
Repeat the last three steps to remove the other intake valve assembly.



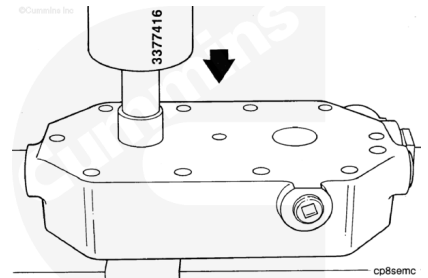
Remove the head.



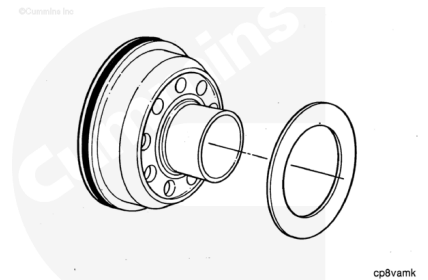
Remove and discard the head gasket.



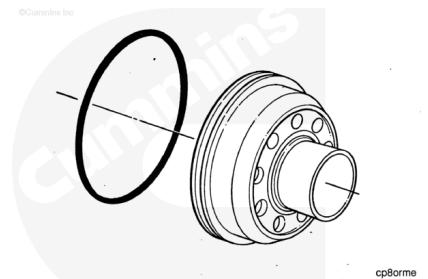
Use a press and an air compressor seat removal tool, Part Number 3377416, to remove the exhaust valve seat assembly from the head.



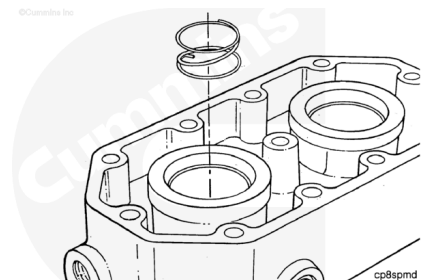
Remove the exhaust valve.



Remove and discard the o-ring seal.

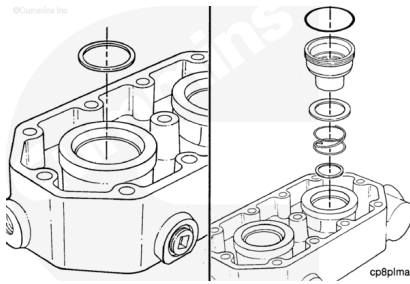


Remove the exhaust valve spring.



Remove the wear plate.

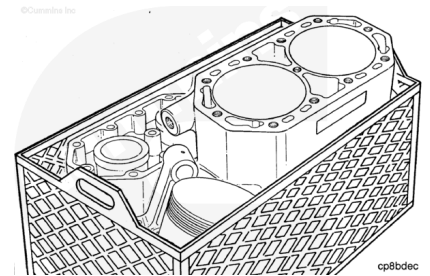
Repeat the last five steps to disassemble the other exhaust valve seat assembly.



Clean and Inspect for Reuse

⚠ WARNING ⚠

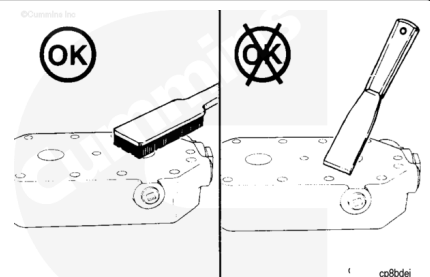
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing.



Soak the parts in a kerosene emulsion based cleaner designed to remove carbon. The cleaner **must** have a pH of 9.5 or less to avoid turning aluminum parts black. The cleaner manufacturer or supplier **must** be contacted about solution concentration, temperature and soak time.

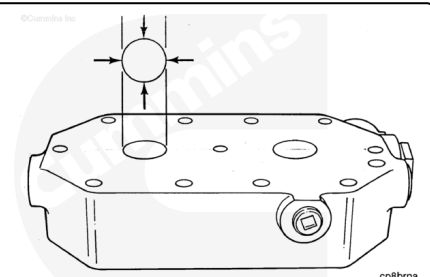
Note : Do not use a scraper to remove carbon and scale, the sealing surfaces can be damaged.

The parts can be scrubbed with a stiff nonmetallic bristle brush.



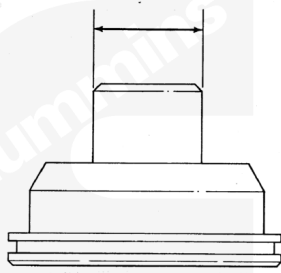
Measure the exhaust valve seat press fit diameter in the cylinder head.

Replace if worn larger than 20.688 mm [0.8145 inch].



Measure the exhaust valve seat press fit diameter.

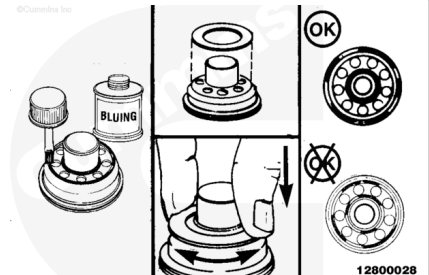
Replace if worn smaller than 20.714 mm [0.8155 inch].



cp8sena

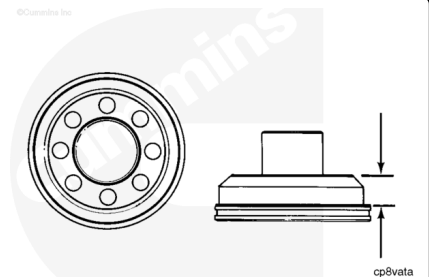
Apply "bluing" to the exhaust seating surface to check the seat.

If the seating surface is **not** 100 percent true discard and replace the valve seat.



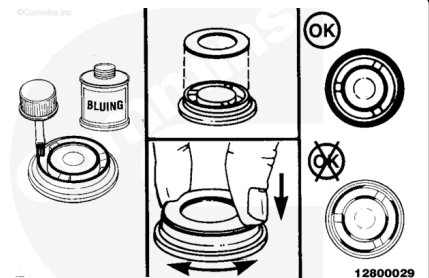
Measure the exhaust valve seat height.

If the height is less than 12.32 mm [0.485 inch] replace the exhaust valve seat.



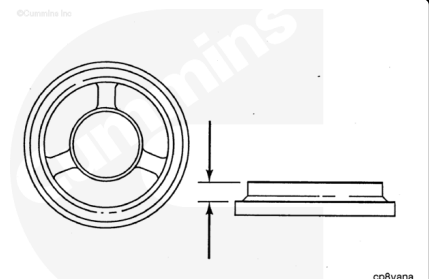
Apply "bluing" to the intake valve seating surface.

If the seating surface is **not** 100 percent true replace the intake valve seat.



Measure the intake valve seat height.

If the height is less than 6.86 mm [0.270 inch] replace the intake valve seat.

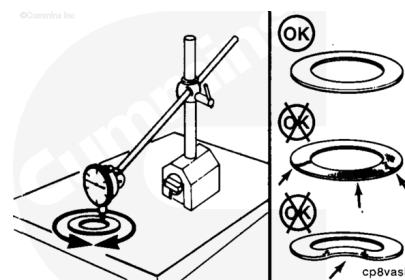


Inspect the exhaust and intake valves for cracks and damage.

Measure valves for flatness. The valves **must** be flat within 0.03 mm [0.001 inch].

Replace the valves if cracked, damaged, or **not** flat.

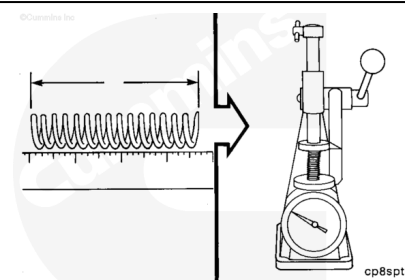
Note : Cummins Turbo Technologies recommends that new valves be installed during rebuild.



Use a valve spring tester, Part Number 3375182, to check the springs.

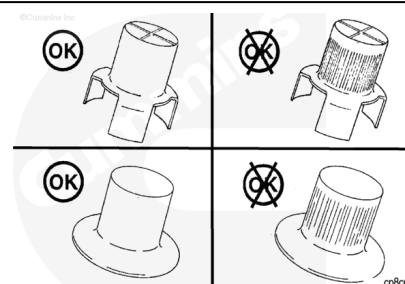
Replace if **not** within limits listed in the Compressor Spring Force Specifications Table at the front of this section.

Note : Cummins Turbo Technologies recommends that new springs be installed during rebuild.



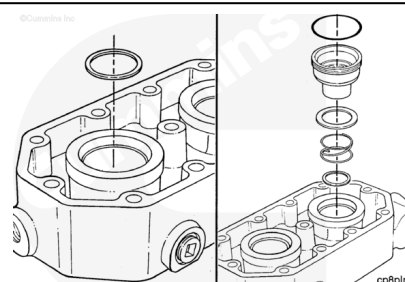
Inspect the upper part of the unloader cap where the rectangular ring seal seats for scoring.

Replace if scored.



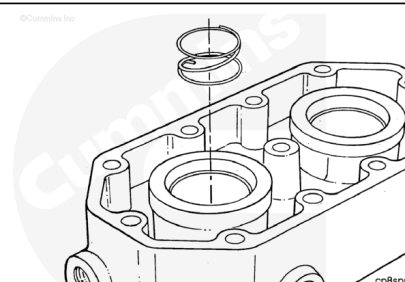
Assemble

Install the wear plate.

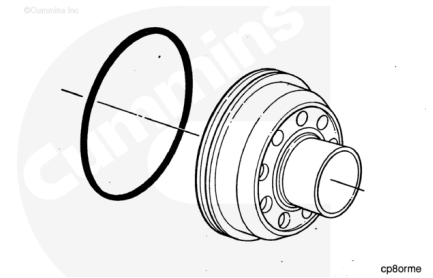


Install the exhaust valve spring with the tang end **down**.

Note : Cummins Turbo Technologies recommends that new springs be installed during rebuild.

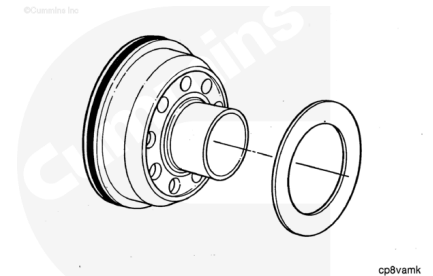


Install a new o-ring seal.

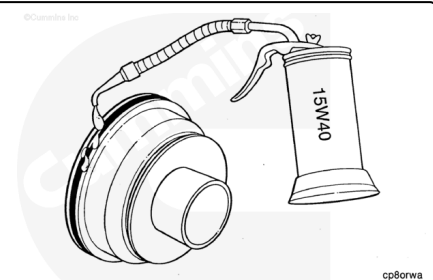


Install the exhaust valve.

Note : Cummins Turbo Technologies recommends that new exhaust valves be installed during rebuild.



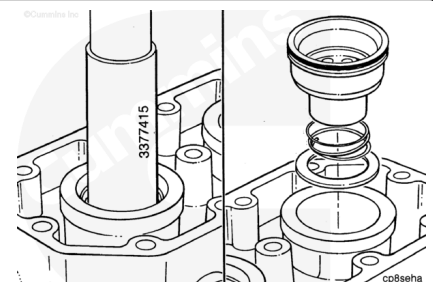
Use clean 15W-40 oil to lubricate the seal.



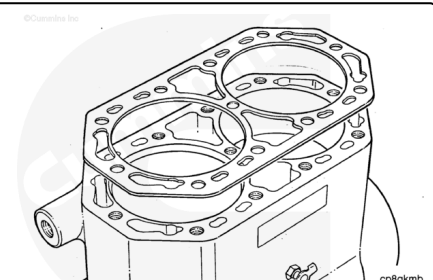
Note : Do not use excessive pressure on the exhaust valve seats, to do so can distort the valve.

Use a hand press and an air compressor seat installation tool, Part Number 3377415, to press the exhaust valve seat into the head.

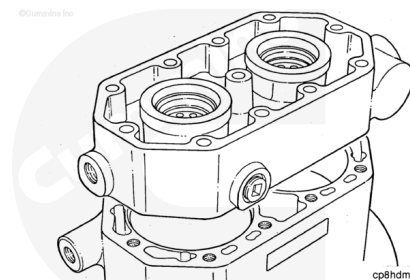
Repeat the last six steps to install the other exhaust valve assembly.



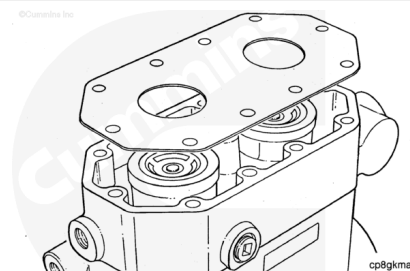
Install a new cylinder head gasket.



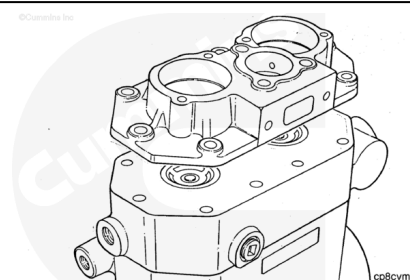
Install the head.



Install a new cover gasket.

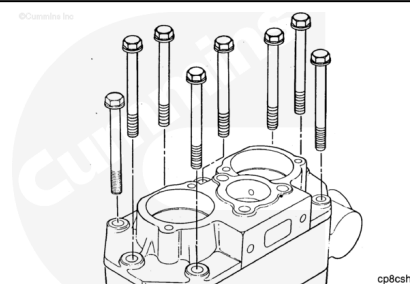


Install the cover.

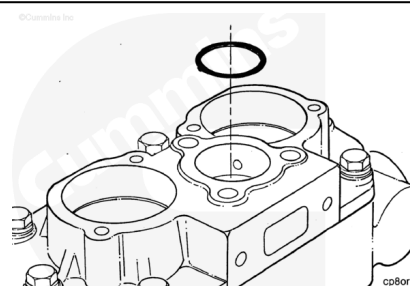


For the ST676, install the eight hexagon head capscrews and finger tighten.

For the ST773, install the eight captive washer capscrews.

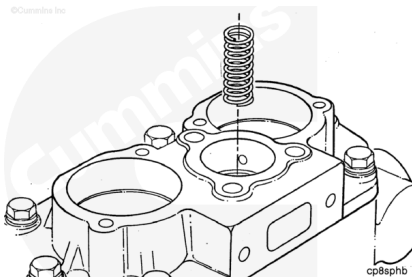


Use an o-ring pick to correctly install the new o-ring seal.



Install the unloading spring.

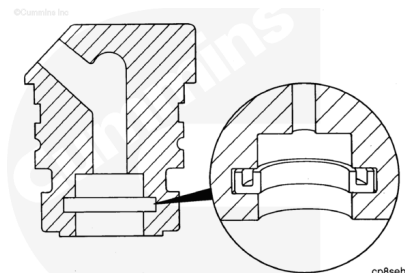
Note : Cummins Turbo Technologies recommends that new springs be installed during rebuild.



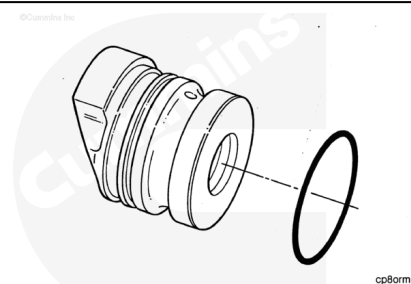
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake failure.

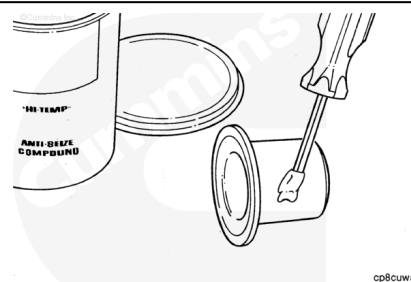
Use an o-ring pick to correctly install the new rectangular ring seal, if the unloader body has a cavity for this o-ring.



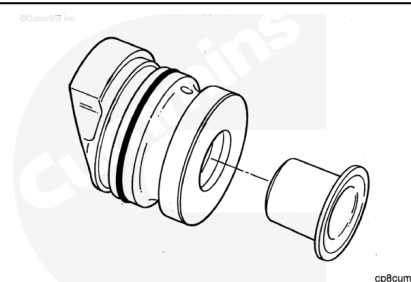
Install a new o-ring seal.



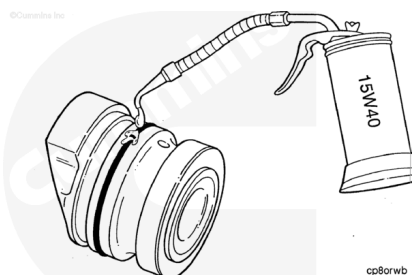
Use antiseize compound to lubricate the outside diameter of the unloader valve cap.



Install the cap into the unloading valve body.



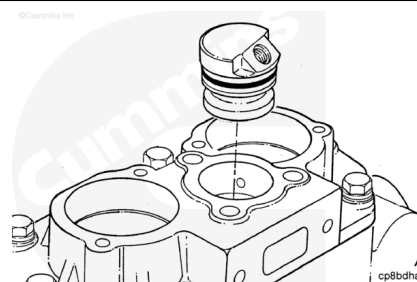
Use clean 15W-40 oil to lubricate the seal.



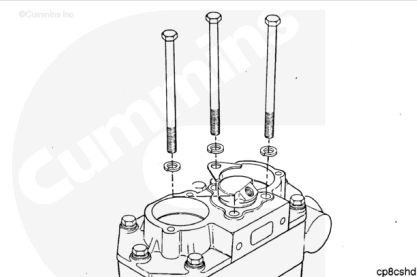
⚠ CAUTION ⚠

The unloader valve body is installed under spring tension. Carefully install to avoid personal injury. Wear face and eye protection.

Install the body into the cover.

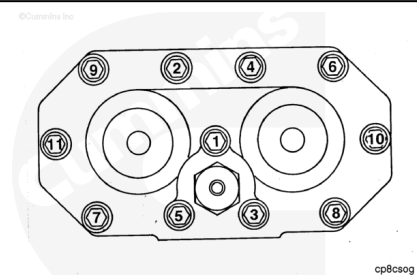


Install the retaining clamp and the three lock washers and the three hexagon head capscrews.



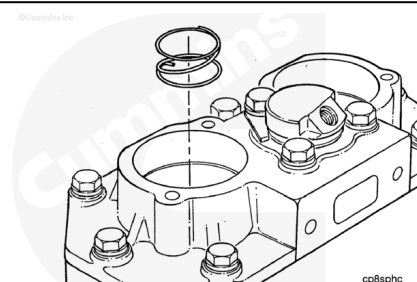
Tighten the capscrews in the sequence shown.

Torque Value: 41 n•m [124 in-lb]



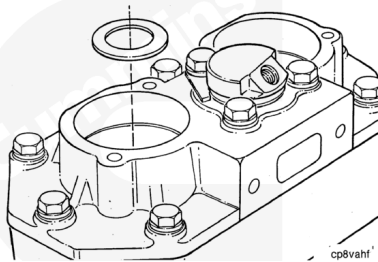
Install the intake valve spring with the tang end **down**.

Note : Cummins Turbo Technologies recommends that new springs be installed during rebuild.

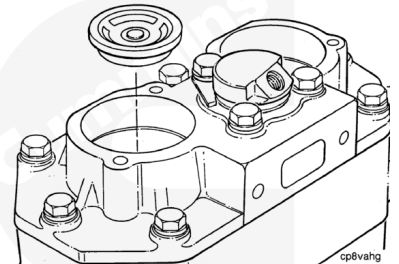


Install the intake valve.

Note : Cummins Turbo Technologies recommends that new intake valves be installed during rebuild.

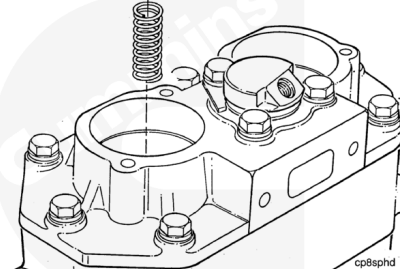


Install the intake valve seat with the flange end **up**.

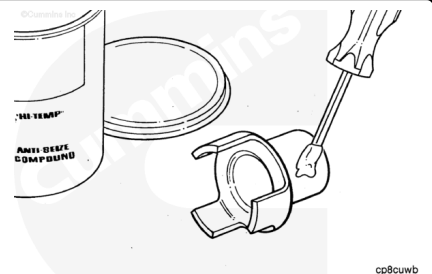


Install the unloader cap spring.

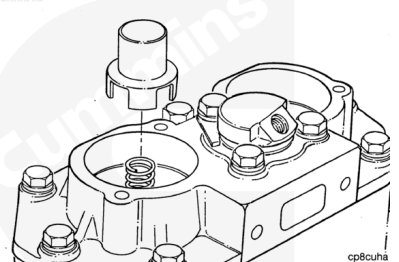
Note : Cummins Turbo Technologies recommends that new springs be installed during rebuild.



Use antiseize compound to lubricate the outside diameter of the unloader valve cap.

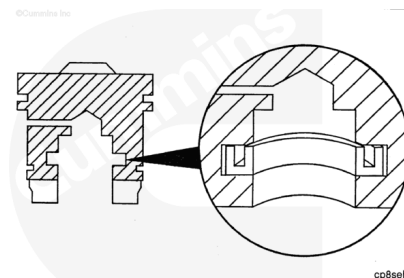


Install the cap into the cover and make sure the three tangs are in the three slots of the intake valve seat.



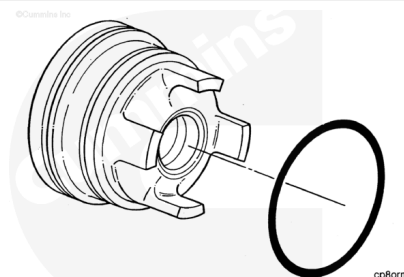
⚠ CAUTION ⚠

The rectangular ring seal must be installed with the grooved side up; failure to do so will result in air system damage and brake failure.

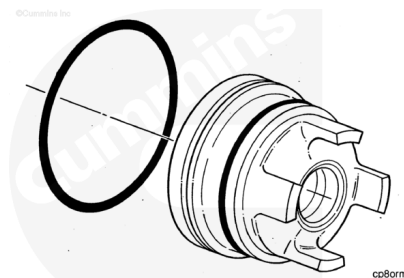


Use an o-ring pick to correctly install the new rectangular ring seal.

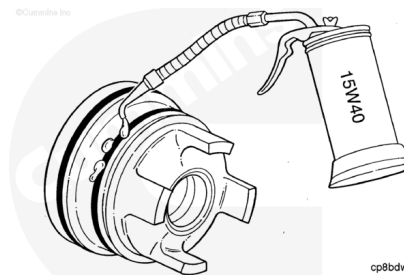
Install a new bottom o-ring seal.



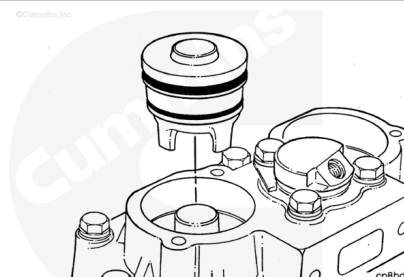
Install a new top o-ring seal.



Use clean 15W-40 oil to lubricate the seals.



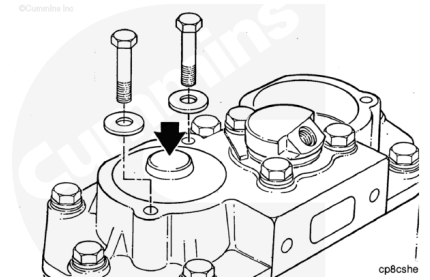
Install the unloading body into the cover.



⚠ WARNING ⚠

The unloader body is installed with spring tension. It must be installed as described to prevent personal injury.

Hold down on the unloading body firmly and install the capscrews and plain washers so as to prevent the unloader body from being thrown free and causing personal injury.



Tighten the capscrews.

Torque Value: 14 n•m [124 in-lb]

Repeat the last thirteen steps to install the other unloading body assembly.

